

Viet-Sang Nguyen

(+33) 6 42 31 73 26
https://nvietsang.github.io

nvietsang@gmail.com
nvietsang

35000 Rennes
nvietsang

Experience

Cadence / Secure-IC Cryptography Engineer

Rennes, France
11/2025–present

- Analyzed potential attacks against classical and post-quantum cryptography and recommended countermeasures (e.g., masking, shuffling, redundancy).
- Assessed and supported cryptographic implementations to meet certification requirements (FIPS, OSCCA, AIS, SESIP, CC).
- Modeled TRNG and analyzed its compliance with different standards.

University of Saint-Étienne Doctoral Researcher

Saint-Étienne, France
01/2023–10/2025

Thesis: Physical Attacks against Symmetric Cryptography Standards 📄🖥️🔒

- Initiated, led, and completed research projects resulting in 5 publications on side-channel and fault attacks.
- Developed and experimentally validated cryptographic attacks; designed and implemented effective countermeasures.
- Excellent grade in Post-Quantum Cryptography Engineering course (awarded 5 bug-bounty points for identifying errors in lectures). 🏆

CryptoExperts

Cryptography Engineer

Paris, France
09/2021–12/2022

- Designed and developed a generic framework for building circuits and generating white-box implementations of cryptographic algorithms. 📄🖥️
- Designed and developed a framework for software obfuscation with a simple stateless secure element. 🔒

CryptoExperts

Cryptography Intern

Paris, France
03/2021–08/2021

- Designed and developed of a cryptocurrency wallet with consideration of security protection. 📄🖥️
- Studied white-box cryptography and physical attacks on ECDSA.

Research Institute XLIM

Cryptography Intern

Limoges, France
06/2020–08/2020

- Studied and developed an encrypted matching scheme. 📄
- Improved speed by pre-discarding costly operations using Cuckoo filter.

Teaching

Embedded Programming (tutorials)

27h×3, for undergraduate students

IUT Saint-Etienne, France
2023–2024

C++ Programming (tutorials)

33h, for undergraduate students

IUT Saint-Etienne, France
2024

Publications

- **Nguyen**, Grosso and Cayrel, “*Practical Persistent Fault Attacks on AES with Instruction Skip*”, IACR CiC, 2025 📄🔒
- **Nguyen**, Grosso and Cayrel, “*Practical Second-Order CPA Attack on Ascon with Proper Selection Function*”, CASCADE, 2025 📄🖥️🔒
- **Nguyen**, Grosso and Cayrel, “*Correlation Power Analysis on Ascon with Multi-bit Selection Function*”, SECRIPT, 2025 📄🖥️🔒
- **Nguyen**, Grosso and Cayrel, “*SIFA on Nonce-based Authenticated Encryption: When Does It Fail? Application to Ascon*”, FDTC, 2025 📄🖥️
- Mercadier, **Nguyen**, Rivain and Udovenko, “*Versatile Software Obfuscation from a Lightweight Secure Element*”, IACR TCHES, 2024 📄🔒
- **Nguyen**, Grosso and Cayrel, “*Provable Resistance of Threshold Implementations to S(I)FA*”, in submission 📄

About Me

My main interest is cryptography engineering. I'm especially motivated by the cycle: analyzing how implementations can be attacked and using those insights to design and implement protections.

Education

PhD in Cryptography

2023–2025

University of Saint-Étienne France
Supervisors: Vincent Grosso, Pierre-Louis Cayrel

Master in InfoSec & Crypto

2019–2021

University of Limoges France
Magna cum laude

Bachelor of Engineering (CS)

2015–2019

HCM University of Technology Vietnam
Magna cum laude, honors program

Scientific Reviewing

External reviewer for EUROCRYPT (2024,2025), TCHES (2026);
Artifact PC member for TCHES (2026)

Honors/Awards

- 2025 1st Prize in NSUCRYPTO Olympiad 🏆
- 2024 1st Prize in NSUCRYPTO Olympiad 🏆
- 2019 Scholarship for 2 years of Master
- 2018 Award for Outstanding Students

Personal Projects

- 2026 Simulation of EUCLEAK attack on ECDSA (2 weeks) 🔒
- 2026 Implementation of ML-KEM in Rust (2 weeks) 🔒; Simulation of CPA attack on ML-KEM (2 weeks) 🔒
- 2025 Threshold Implementation of PRESENT cipher in ARM (2 weeks) 🔒
- 2021 Tutorials for DPA attacks (1 weeks) 🔒

Technical Skills

- ★ Skilled in both research and engineering.
- 🔒 – Cryptography: symmetric, asymmetric, post-quantum, protocols
- Development: implementation, evaluation
- Attacks: side-channel, fault injection
- Countermeasure: design, analysis
- Standards and certification compliance
- Reverse engineering
- 🔗 Python, C/C++, Assembly, Rust, GPGPU
- 🔗 Sagemath, ChipWhisperer, SCALib, Side-Channel Marvels

Languages

English Full professional proficiency
French Level B2
Vietnamese Native

Reference

Dr. Vincent Grosso

CNRS Researcher at University of Saint-Étienne
18 rue Prof. Benoît Lauras, Saint-Étienne
vincent.grosso@cnrs.fr